

ReDataX

M1: Multiscale Model

Egor Vasilev

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1 Model role and testable hypothesis

M1 is the first extension of the baseline M0 model. M0 describes order flow at a single scale; M1 looks at the same flow across several scales. Let us introduce the following definition.

Definition 1.1. The *information set* $\mathcal{I}_t$ is the collection of data available no later than the decision time. A feature is admissible if it is measurable with respect to $\mathcal{I}_t$.

The only difference between the two models being compared is therefore their information set. Everything else - the target variable, forecast horizon, classification algorithm, temporal splits, and metrics - remains unchanged. This gives us the following hypothesis.

Hypothesis 1.1. A multiscale representation of the signed flow of a single market contains more useful information about its state than a single-scale approximation. In other words, for the same label $Y_{t,H}$ and the same general feature-construction procedure,

$$\mathbb{P}(Y_{t,H} = 1 \mid X_t^{(1)}) \neq \mathbb{P}(Y_{t,H} = 1 \mid X_t^{(0)}),$$

where $X_t^{(0)}$ and $X_t^{(1)}$ are the M0 and M1 feature vectors, respectively. M1 is accepted if the difference in Average Precision, averaged across days, is positive.

2 Why a multiscale description is useful

The same imbalance measured over one window can correspond to very different market states: a short impulse against a neutral long-run background, a persistent one-way flow across all scales, a reversal with opposite signs on short and long windows, or one large trade unsupported by the rest of the flow. M0 cannot distinguish these cases when the selected single-scale statistic happens to be the same. M1 instead describes the current market state by a vector of the same statistic evaluated over a family of nested windows.

Definition 2.1. A *scale family* $\mathcal{B}$ is an ordered set of time intervals selected for feature construction.

Definition 2.2. Let $\mathcal{B} = \{\tau_1, \tau_2, \dots, \tau_K\}$ be a scale family, and let $f^{(\tau)}$ be some flow statistic evaluated over the window $(t - \tau, t]$. The *profile* of f at time t is the vector

$$\mathbf{f}_t = (f_t^{(\tau_1)}, f_t^{(\tau_2)}, \dots, f_t^{(\tau_K)})^\top.$$

For example, a volume-imbalance profile can separate a noisy spike from an emerging trend and can catch a sign reversal before it becomes visible in a single window. Short scales pick up fast changes; long scales retain the accumulated background. An analogy with statistical physics is useful here. In a many-particle system, fluctuations may be short-wavelength or long-wavelength. The former appear over small distances and usually remain local, whereas the latter can reshape the system as a whole. A familiar example is a smooth stream of water from a tap that eventually breaks up under long-wavelength perturbations.

3 Data and the signed-flow field

The source events are Binance Spot aggregate trades (`aggTrades`). Each trade k is described by its price, notional n_k in the quote currency, timestamp, and the `m` flag. The flag is `true` when the buyer was the maker and `false` otherwise. The ReDataX parser assigns the aggressor-side sign ε_k as

$$\varepsilon_k = \begin{cases} +1, & \text{m=false} \quad (\text{aggressive purchase of the base asset}), \\ -1, & \text{m=true} \quad (\text{aggressive sale of the base asset}). \end{cases}$$

Definition 3.1. For an arbitrary set of events W , the *signed flow* is

$$\phi(W) = \frac{\sum_{k \in W} \varepsilon_k n_k}{\sum_{k \in W} n_k},$$

where n_k is the trade notional in the quote currency. If the denominator is positive, then $\phi(W) \in [-1, 1]$.

At every scale τ_i , trades are first aggregated into whole-second buckets. For target market A , aggressive buy and sell volumes $V_A^+(t)$ and $V_A^-(t)$ are calculated for every completed second t . The one-second imbalance is then

$$\phi_A(t) = \begin{cases} \frac{V_A^+(t) - V_A^-(t)}{V_A^+(t) + V_A^-(t)}, & V_A^+(t) + V_A^-(t) > 0, \\ 0, & \text{when no trades occur.} \end{cases}$$

For each scale $B \in \mathcal{B}$, define the smoothed field

$$\bar{\phi}_A^{(B)}(t) = \frac{1}{B} \sum_{u=0}^{B-1} \phi_A(t-u).$$

This is a moving average over calendar seconds, including the current completed second. Seconds without trades remain in the denominator B , which reduces the average field amplitude during quiet periods.

4 Model features

Four groups of features are constructed from the multiscale field $\bar{\phi}_A^{(B)}(t)$ and the current flow direction $d_A(t) = \text{sign } \phi_A(t)$.

Definition 4.1. At scale B , the *aligned flow* is

$$h_{A,B}(t) = d_A(t) \bar{\phi}_A^{(B)}(t).$$

A positive value means that the historical flow at this scale points in the same direction as the current aggregate flow.

Definition 4.2. The *imbalance amplitude* is the absolute value of the smoothed field,

$$r_{A,B}(t) = |\bar{\phi}_A^{(B)}(t)|.$$

In plain terms, it measures how unbalanced the flow is, regardless of direction.

To let a linear model distinguish weak from strong imbalances independently of their sign, we add even powers of the smoothed field:

$$a_{A,B}(t) = \left(\bar{\phi}_A^{(B)}(t)\right)^2, \quad b_{A,B}(t) = \left(\bar{\phi}_A^{(B)}(t)\right)^4.$$

Critical-phenomena theory and quantum field theory contain a model commonly known as the ϕ^4 model:

$$\mathcal{H} = \int d^D x \left[\frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 \right]. \quad (1)$$

This Hamiltonian describes a system near a phase transition. The quadratic term ϕ^2 represents the energetic cost of field fluctuations, while ϕ^4 represents self-interaction and helps determine the nature of the transition. This analogy motivates the use of even powers of the imbalance field here. It is a feature-engineering argument, not a claim that the financial system literally follows a ϕ^4 field theory.

Finally, the absolute size of the current second is recorded through the logarithm of total notional:

$$\ell_A(t) = \log(1 + V_A^+(t) + V_A^-(t)).$$

For every admissible time t , the resulting feature vector is

$$X^{\text{M1}}(t) = \left[\ell_A(t), \{h_{A,B}(t)\}_{B \in \mathcal{B}}, \{r_{A,B}(t)\}_{B \in \mathcal{B}}, \{a_{A,B}(t)\}_{B \in \mathcal{B}}, \{b_{A,B}(t)\}_{B \in \mathcal{B}} \right]^\top,$$

with dimension

$$\dim X^{\text{M1}} = 1 + 4|\mathcal{B}| = 29. \quad (2)$$

5 Target variable

For the one-second implementation, let $P_A(t)$ denote the volume-weighted average price of target market A during completed second t . The aggressor-aligned markout at horizon H , measured in basis points, is

$$m_{A,H}(t) = 10^4 d_A(t) \frac{P_A(t+H) - P_A(t)}{P_A(t)},$$

where $d_A(t) = \text{sign } \phi_A(t)$. The binary label equals one when the markout is positive:

$$Y_{A,H}(t) = \mathbf{1}\{m_{A,H}(t) > 0\}.$$

An observation is admissible only when the target market is active during second t , $d_A(t) \neq 0$, and VWAP exists exactly at both t and $t+H$. The last observed price is not carried forward. The final cross-market experiment fixes the horizon at $H = 5$ seconds.

A positive markout means that the price moved in the direction of the current aggregate flow. This creates adverse-selection risk for a liquidity provider that took the opposite side.

6 Statistical model and training

Like M0, M1 uses logistic regression. After fitting feature standardization on the training sample, the probability of an adverse outcome is parameterized by a sigmoid:

$$\hat{p}_t = \sigma(\beta_0 + \beta^\top z_t), \quad \sigma(u) = \frac{1}{1 + e^{-u}}.$$

The parameter vector β is estimated by minimizing

$$\mathcal{L}(\beta) = - \sum_{i \in \text{train}} [Y_i \log p_i + (1 - Y_i) \log(1 - p_i)] + \lambda \|\beta\|_2^2.$$

The implementation uses scikit-learn’s `SGDClassifier` with logistic loss, L_2 regularization, a `StandardScaler`, averaged coefficients, and a fixed random-number generator.

Nested time windows naturally create correlation between features at adjacent scales because wider smoothed fields inherit information from narrower windows. Without regularization, this is an easy route to overfitting. Regularization reduces variance and improves out-of-sample behavior, but it also means that an individual coefficient - for example, the coefficient of aligned flow at scale B - should never be interpreted without accounting for correlation with neighboring scales.

In the one-second implementation (REAL-05), the candidates $\{10^{-5}, 10^{-4}, 10^{-3}\}$ are compared using mean daily AP. The final experiment selects $\alpha = 10^{-3}$ for all three models. Once the hyperparameter is chosen, both the model and the scaler are refitted on the combined training and development samples.

7 Temporal validation and metrics

The primary protocol does not shuffle individual observations across calendar periods. For the M1-versus-M0 comparison, the event-level implementation uses the following intervals: training from 6 January to 15 January 2025, development from 16 January to 19 January 2025, and testing from 20 January to 26 January 2025.

The reported metrics are ROC-AUC, Average Precision (AP), the Brier score, and positive-class lift in the top 10% of predictions. Uncertainty in the AP difference is estimated by resampling calendar days with replacement using 5,000 bootstrap replications. The interval captures variation across days; with only seven test days, however, it should be read as a limited robustness estimate rather than an asymptotic result.

8 Results against M0

Table 1 reports the event-level M1-versus-M0 results on the REAL-02 test. For both markets and both horizons, the daily AP difference is positive on all seven test days.

The evidence supports the claim: at short horizons, a multiscale flow representation ranks positive aggressor-aligned markouts better than the fixed-scale M0 baseline. This does not mean that M1 predicts medium-term exchange rates, guarantees realized profit, or reconstructs the hidden state of a particular market maker. The one-second M1 specification fixed in REAL-05 is the baseline for M2 and M3. Its absolute metrics are recorded

Table 1: Daily AP difference ($M1 - M0$) on the REAL-02 test.

Market	H , s	mean ΔAP	95% bootstrap interval
BTCUSDT	1	0.01960	[0.01423; 0.02484]
BTCUSDT	5	0.01727	[0.01095; 0.02406]
ETHUSDT	1	0.01827	[0.01272; 0.02353]
ETHUSDT	5	0.01106	[0.00597; 0.01647]

in the experiment registry and should not be compared mechanically with those of the event-level implementation.

9 Limitations and reproducibility

M1 inherits all M0 limitations and adds several of its own. Adjacent scales reuse a large share of the same trades, creating correlated features and calling for caution when individual coefficients are interpreted. The fixed grid $\mathcal{B}$ need not be optimal for every market: windows with the same duration may contain very different numbers of events as trading intensity changes.

Features from different scales enter additively; the model does not include explicit scale derivatives or ratios. Its information set is still limited to a single market, so the state of related trading pairs is ignored. Finally, M1 is a binary classifier and does not distinguish movements by magnitude.

Historical `aggTrades` do not contain the order book, fees, position ownership, or actual execution costs. The target is built from future VWAP and is useful for ranking research, but it is not identical to net P&L. Seven test days are not enough to establish robustness across all market regimes.

Finally, using several scales does not prove scale invariance and does not imply the existence of a renormalization-group flow in the financial market.

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